DBMS - PROJECT

Q Advanced Online Quiz Platform

Quiz Made Easy

A Secure, Role-Based Quiz Platform for Academic & Webinar Assessments

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Dept. of CSE — B.Tech (Cyber Security) · April 2026

Second Year/ Fourth Semester

4 User Roles	**2** Quiz Modes
11 DB Tables	**Live** Deployed

qez.vercel.app

quizzkr.vercel.app

Built with Next.js · PostgreSQL · Prisma · Vercel

Advanced Online Quiz Platform

What is Qez?

A full-stack, database-driven quiz platform supporting academic exams and live webinar competitions — in one unified system.

qez.vercel.app

quizzkr.vercel.app

Introduction

Why Choose Us?

Academic Mode

Timed, structured exams for classroom use with anti-cheat monitoring and detailed analytics.

Webinar Mode

Live competitive quiz events with real-time leaderboards and speed-based ranking.

Anti-Cheat

Tab-switch, right-click, DevTools and fullscreen detection — logged for review.

DBMS First

11 tables, 8 enums, full FK constraints, JSONB drafts, strategic indexing.

Problem Statement

What's Broken Today?

No Role-Based Architecture

Platforms treat all users equally — no distinction between teachers, students, hosts, or admins.

Client-Side Score Submission

Scores sent from the frontend are trivially manipulable via browser DevTools.

No Anti-Cheat Infrastructure

Tab switches, copy attempts, and DevTools usage go undetected and unlogged.

No Unified Dual-Mode System

Academic and live competition quizzes need different logic — yet no platform unifies both.

Weak Authentication

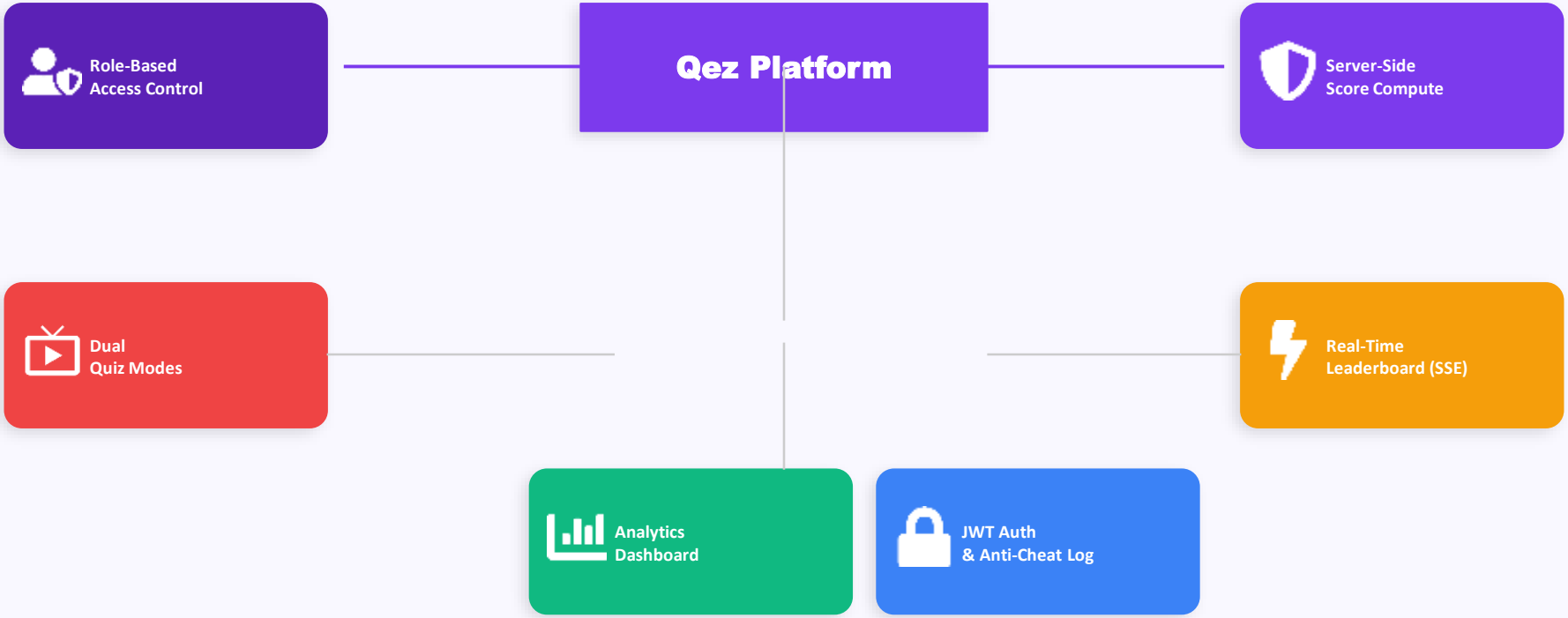
Many quiz tools lack email verification, OAuth flows, and CAPTCHA protection.

Poor Database Design

Flat data structures, no normalization, weak referential integrity — leading to inconsistencies.

Proposed Solution

How We Solves It



Platform Features

Key Features

01

Timed MCQ Questions

Per-question countdown with auto-advance and server-side time tracking.

02

Draft Answer Recovery

JSONB columns save answers every keystroke — quiz survives page refresh.

03

Server-Side Scoring

Frontend never submits a score — only option keys. Server evaluates correctness.

04

Anti-Cheat Monitoring

Logs tab switches, right-clicks, DevTools attempts, fullscreen exits with warning levels.

05

Real-Time Leaderboard

SSE-powered live rankings sorted by score DESC, time ASC — updates as users submit.

06

Analytics & Certificates

Per-question accuracy, score distributions, suspicious attempt review, PDF certificates.

Access Control

System Roles & Permissions

Role-based access enforced at JWT payload, API route handlers, and database ownership layers.

STUDENT

- Join quizzes via 6-char code
- Attempt timed MCQ questions
- View leaderboard (if enabled)
- Download completion certificates
- Track quiz attempt history

TEACHER

- Create & manage academic quizzes
- Add questions with difficulty levels
- View analytics dashboard
- Export results as CSV
- Review suspicious attempts

WEBINAR_HOST

- Host live competitive events
- Manage real-time leaderboard
- Award points & ranks
- Export winner certificates
- Control leaderboard visibility

ADMIN

- Full platform oversight
- View all users & quizzes
- Access admin dashboard
- Monitor system-wide activity
- Bypass standard login flow

Dual-Mode Architecture

Two Modes, One Platform

ACADEMIC MODE

For institutional, structured exams

- Teacher-controlled private assessment
- Per-question timed countdown
- Private results until released
- Anti-cheat warnings with warning levels
- Analytics accessible to teacher only
- Shuffle questions & options support
- CSV export of all attempts

WEBINAR MODE

For live competitive quiz events

- Webinar Host–controlled live competition
- Shared real-time timing for all
- Public leaderboard with SSE updates
- Speed-based ranking (score → time)
- Points awarded per participant rank
- Winner CSV and certificate export
- Configurable leaderboard visibility

System Architecture

Full-Stack Architecture Overview

User (Browser)

Chrome / Firefox / Safari

Presentation Layer

Next.js 15 · React 19 · Tailwind CSS · Framer Motion

Application Layer

Next.js API Route Handlers · Zod Validation · JWT Auth · bcrypt

Data Layer

Prisma ORM · PostgreSQL on Neon DB (serverless, auto-scaling)

DevOps & Infra

Vercel Deployment · Cloudflare Turnstile · Gmail SMTP · Google OAuth

Technology Stack

Built With Modern Tools

Frontend

- Next.js 15 (App Router)
- React 19
- TypeScript
- Tailwind CSS
- Framer Motion

Backend

- Next.js Route Handlers
- Zod Validation
- jsonwebtoken (JWT)
- bcryptjs (10 rounds)
- Nodemailer (SMTP)

Database

- PostgreSQL (Neon DB)
- Prisma ORM
- JSONB columns
- Enums & Indexes
- Version-controlled migrations

DevOps & Auth

- Vercel (CI/CD)
- Google OAuth 2.0
- Cloudflare Turnstile
- Gmail SMTP
- GitHub

Database Design

PostgreSQL Schema — 11 Tables

User

Stores all roles, credentials, OAuth data

Quiz

Mode, state, join code, scheduling config

Question

Prompt, difficulty, time limit, correct key

QuestionOption

4 MCQ choices per question (A/B/C/D)

Attempt

Per-user attempt with draft JSONB columns

Response

Per-question answer, correctness, time taken

SuspiciousEvent

Anti-cheat log (tab switch, right-click…)

WebinarResult

Rank and points for webinar participants

CertificateClaim

Certificate claims per user per quiz

PasswordResetToken

Hashed reset tokens with expiry

EmailVerificationToken

Email activation tokens

Key Constraints: Unique(email, joinCode, userId+quizId, attemptId+questionId) · FK Cascades on delete · JSONB for draft answers · 8 PostgreSQL Enums · Strategic indexing on FK & filter columns

Integrity System

Anti-Cheat Monitoring

Tab Switch

TAB_SWITCH

Detected when user leaves the quiz tab. Each switch logged with timestamp.

Right-Click

RIGHT_CLICK

Context menu is disabled. Any attempt is captured as a suspicious event.

DevTools Open

DEVTOOLS_ATTEMPT

Keyboard shortcuts like F12, Ctrl+Shift+I trigger a warning and log.

Fullscreen Exit

FULLSCREEN_EXIT

Leaving fullscreen during a quiz attempt is flagged and recorded.

Copy Attempt

COPY_ATTEMPT

Ctrl+C and text selection are blocked and logged.

Window Blur

OTHER

Focus leaving the quiz window raises warning level immediately.

⚠ **We prioritize fairness: suspicious behavior is LOGGED FOR REVIEW — never auto-disqualifies. Teachers decide.**

User Journey

Quiz Attempt Workflow

1. **Login & Verify**
   JWT issued after email verification
2. **Join via Code**
   6-char join code validates quiz state & eligibility
3. **Attempt Questions**
   Per-question countdown, drafts saved to JSONB in real time
4. **Anti-Cheat Active**
   Client logs suspicious events → server stores in SuspiciousEvent table
5. **Submit / Auto-Submit**
   Option keys sent to server → score computed server-side
6. **Results & Leaderboard**
   Score, ranking, analytics, certificates available

Draft Recovery: JSONB columns persist answers on every selection — quiz survives accidental page refresh.

Results & Insights

Leaderboard & Analytics

Leaderboard

- Ranked by Score (DESC) → Time (ASC) → Submitted At
- Visibility: HIDDEN · TOP_10 · FULL
- Real-time updates via Server-Sent Events (SSE)
- Webinar points and rank stored in WebinarResult
- Winners CSV export for prize distribution

Analytics

- Total attempts, avg score, avg completion time
- Accuracy by difficulty (Easy / Medium / Hard)
- Per-question accuracy & avg time taken
- Score distribution chart (0–25%, 26–50%, etc.)
- Warning level distribution (Clean / Warned / Flagged)
- Best performer & hardest question callouts

Planned SQL Views: leaderboard_view · quiz_analytics_view · question_analytics_view · student_analytics_view

Roadmap

Future Enhancements

PL/pgSQL Stored Procedures

Move submit_quiz() & update_webinar_rankings() to database layer.

AFTER INSERT Triggers

Auto-update Attempt scores via PostgreSQL triggers on Response inserts.

SQL Views (4 planned)

leaderboard_view, quiz_analytics_view, question_analytics_view, student_analytics_view.

More Question Types

Extend QuestionType enum: True/False, Short Answer, Fill-in-the-Blank.

Mobile App (React Native)

Consume existing Next.js API endpoints in a native mobile experience.

Webcam Proctoring

Integrate third-party proctoring for high-stakes exams.

Advanced Online Quiz Platform

- A unified platform for academic exams and live webinar competitions
- Server-side scoring with full ACID-compliant PostgreSQL persistence
- Anti-cheat monitoring with SuspiciousEvent logging (no auto-disqualify)
- Real-time leaderboard via Server-Sent Events in Webinar Mode
- Role-based access enforced at JWT, API, and database layers
- 11 tables · 8 enums · JSONB drafts · strategic indexes — deployed on Vercel

Thank You & Questions Welcome

Try it live

qez.vercel.app

GitHub

github.com/bit-roop/qez

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